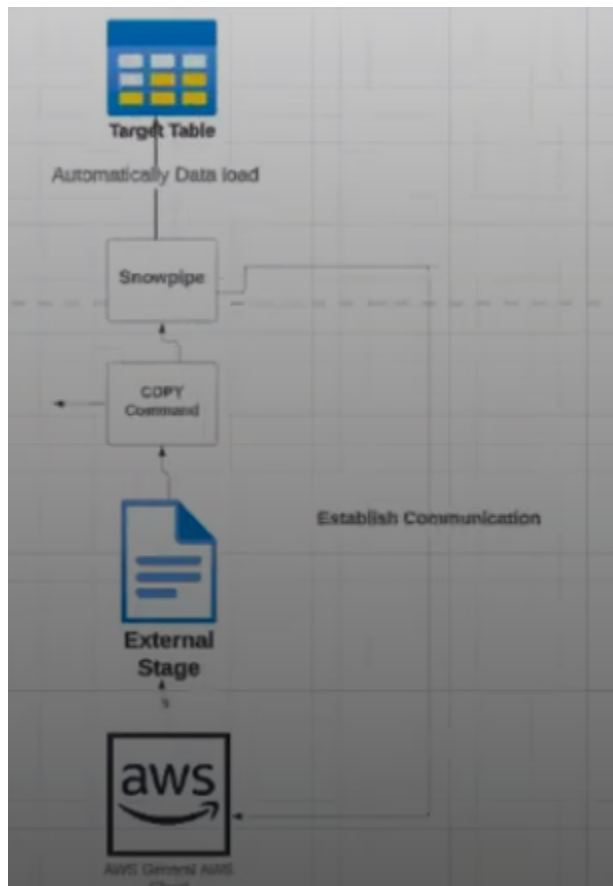


Snow pipe:

Whenever you add data in t s3 ,it wud automatically add data in target table using snow pipe, that's the use of snow pipe



```
CREATE OR REPLACE TABLE employees1 (  
  id INT,  
  first_name STRING,  
  last_name STRING,  
  email STRING,  
  location STRING,  
  department STRING  
);
```

Create a IAM ROLE as below

The screenshot shows the AWS IAM console for the role 'snowflakenewrolefeb2025'. The 'Summary' section displays the role's creation date (February 14, 2025, 20:00 UTC-05:00) and its ARN: `arn:aws:iam::97505034868:role/snowflakenewrolefeb2025`. The 'Last activity' shows it was used 21 minutes ago. The 'Trust relationships' tab is active, showing a trust policy that allows the role to assume the role 'snowflakenewrolefeb2025' in the account '97505034868'.

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Principal": {
7-         "AWS": "arn:aws:iam::97505034868:role/snowflakenewrolefeb2025"
8-       },
9-       "Action": "sts:AssumeRole",
10-      "Condition": {
11-        "StringEquals": {
12-          "sts:ExternalId": "VL834314-KECRole-2025-0719VUVxLG8m4="
13-        }
14-      }
15-    }
16-  ]
17- }
```

And then add s3 full access

Now we have to integrate AWS s3 with Snowflake using below procedure

```
create or replace storage integration s3_int1345
TYPE = EXTERNAL_STAGE
STORAGE_PROVIDER = S3
ENABLED = TRUE
STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::97505034868:role/snowflakenewrolefeb2025'
STORAGE_ALLOWED_LOCATIONS = ('s3://snowflakess3feb2025/snowpipes/')
COMMENT = 'integration object' ;

desc integration s3_int1345;
```

	property	property_type	property_value	property_default
1	ENABLED	Boolean	true	false
2	STORAGE_PROVIDER	String	S3	
3	STORAGE_ALLOWED_LOCATIONS	List	s3://snowflake3feb2025/snowpipes/	[]
4	STORAGE_BLOCKED_LOCATIONS	List		[]
5	STORAGE_AWS_IAM_USER_ARN	String	arn:aws:iam::0477xxxxx:user/a20w0000-s	
6	STORAGE_AWS_ROLE_ARN	String	arn:aws:iam::9750xxxxx:role/snowflakeenwrolefeb2025	
7	STORAGE_AWS_EXTERNAL_ID	String	VLB36724_S5PXXXXX-36G00UO7NYUVxLGBm4=	
8	USE_PRIVATELINK_ENDPOINT	Boolean	false	false
9	COMMENT	String	integration object	

les

> snowflakenewrolefeb2025

Access t (IAM)

<

snowflakenewrolefeb2025

Info

Summary

Creation date
February 14, 2025, 20:00 (UTC-05:00)

ARN
arn:aws:iam:975050348684:role/snowflakenewrolefeb2025

Link to switch roles in console
<https://signin.aws.amazon.com/switchrole?roleName=snowflakenewrolefeb2025&account=975050348684>

Maximum session duration
1 hour

Last activity
21 minutes ago

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Trusted entities

Entities that can assume this role under specified conditions.

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Principal": {
7-         "AWS": "arn:aws:iam::975050348684:user/a20w0000-s"
8-       },
9-       "Action": "sts:AssumeRole",
10-      "Condition": {
11-        "StringEquals": {
12-          "sts:ExternalId": "VIA-975050348684:VIA-975050348684"
13-        }
14-      }
15-    }
16-  ]
17- }
```

LIST @csv_folder123456 ;

Results Chart				
	name	size	md5	last_modified
1	s3://snowflakess3feb2025/snowpipes/employee_data_2.csv	6398	09841c95996ecd4f55e6ce7e1aacc77d	Fri, 21 Feb 2025 22:16:00 GMT
2	s3://snowflakess3feb2025/snowpipes/employee_data_4.csv	6412	ebba26266b70e810d1c629d3221d615b	Mon, 17 Feb 2025 22:44:50 GMT

next is pipe creation

Create pipe and describe pipe as below:

And then copy notification channel in to aws sns queue under s3 properties

```

28 // Define pipe
29 CREATE OR REPLACE pipe employee_pipes
30 auto_ingest = TRUE
31 AS
32 COPY INTO employees1
33 FROM @csv_folder123456 ;
34
35 DESC pipe employee_pipes;
36
37

```

definition	owner	notification_channel	comment	integration
COPY INTO employees1 FROM @csv_folder123456	ACCOUNTADMIN	arn:aws:sqs:us-east-1:047719618228:sf-snowpipe-AIDAQWHCPS22J4ASSBE3R--igwPe		null

Query Details

Query duration 36ms

Rows 1

Query ID 01ba8cc9-030c-7560-0...

Show more

Ask Conint

S3 bucket :

Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)

Server access logging
Disabled

AWS CloudTrail data events [Info](#)

[Configure in CloudTrail](#)

Configure CloudTrail data events to log Amazon S3 object-level API operations in the CloudTrail console. [Learn more](#)

Name

Access

No data events
No data events to display.

[Configure in CloudTrail](#)

Event notifications (1/1)

[Edit](#)[Delete](#)[Create event notification](#)

Send a notification when specific events occur in your bucket. [Learn more](#)

☒	Name	Event types	Filters	Destination type	Destination
☒	snowflakenotification	All object create events	snowpipes/	SQS queue	arn:aws:sqs:us-east-1:123456789012:snowflake-igwPe5ZciF3i6iLrJ4JIA

Amazon EventBridge

[Edit](#)

For additional capabilities, use Amazon EventBridge to build event-driven applications at scale using S3 event notifications. [Learn more](#) or [see EventBridge pricing](#)

Send notifications to Amazon EventBridge for all events in this bucket
Off

Transfer acceleration

[Edit](#)

```
26 truncate employee_pipe;  
27  
28 // Define pipe  
29 CREATE OR REPLACE pipe employee_pipes  
30 auto_ingest = TRUE  
31 AS  
32 COPY INTO employees1  
33 FROM @csv_folder123456 ;  
34  
35 DESC pipe employee_pipes;  
36  
37  
38 truncate table employees1 ;  
39  
40 SELECT * FROM employees1 ;  
41
```

} PIPE

Results							Chart
#	ID	FIRST_NAME	LAST_NAME	EMAIL	LOCATION	DEPARTMENT	Query Details
	1	Claus	Petruska	cpetruska0@nifty.com	Baru	Marketing	Query duration
	2	Tyrus	Searchfield	tsearchfield1@opensource.org	Padabeunghar	Marketing	Rows
	3	Henrie	Boat	hboat2@digg.com	Laibin	Engineering	Query ID